

2026/1/13

Introduction to the RAMPS Project

Christopher Skora



RAMPS Experiment Intro

These precise measurements are useful in fields such as nuclear physics, geochronology, and particle astrophysics

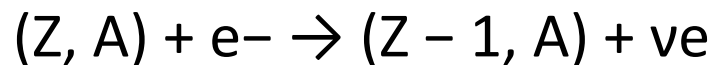
These half life measurements of long-lived isotopes can teach us about electroweak coupling, as well as help in determining factors of $0\nu\beta\beta$, and s-nucleosynthesis in stars

(Creation of new nuclei from existing protons and neutrons)

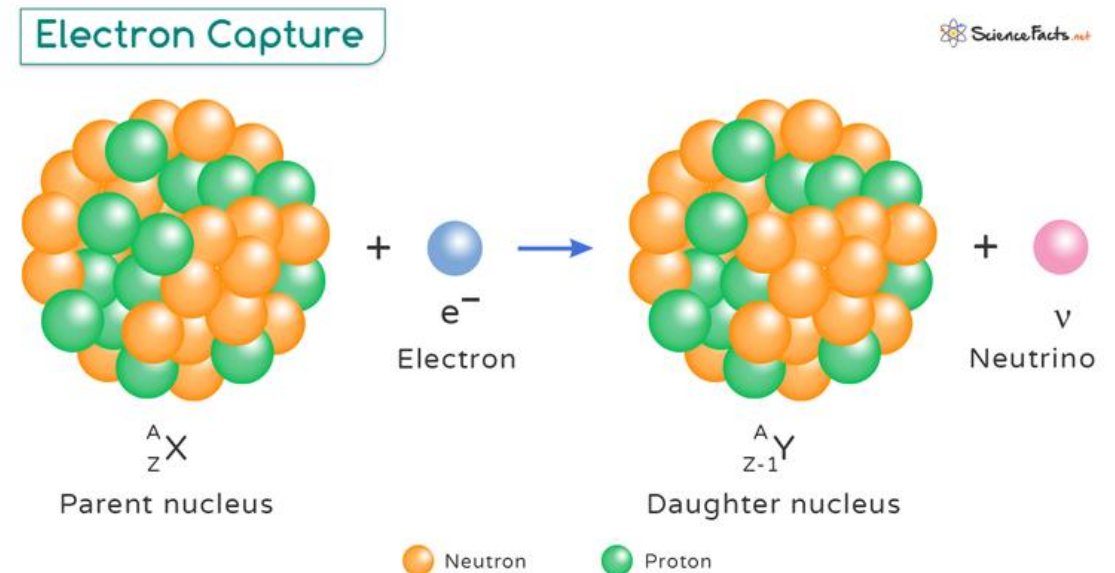
We are looking into an electron capture decay, which only occurs in six known long lived nucleotides, and we are measuring the ^{176}Lu decay, which has never been measured, and is highly forbidden

Electron Capture Decay

- EC is a nuclear process where an atomic electron is captured by the nucleus, interacting with a proton
- Causes a neutron to form alongside an ejected electron neutrino

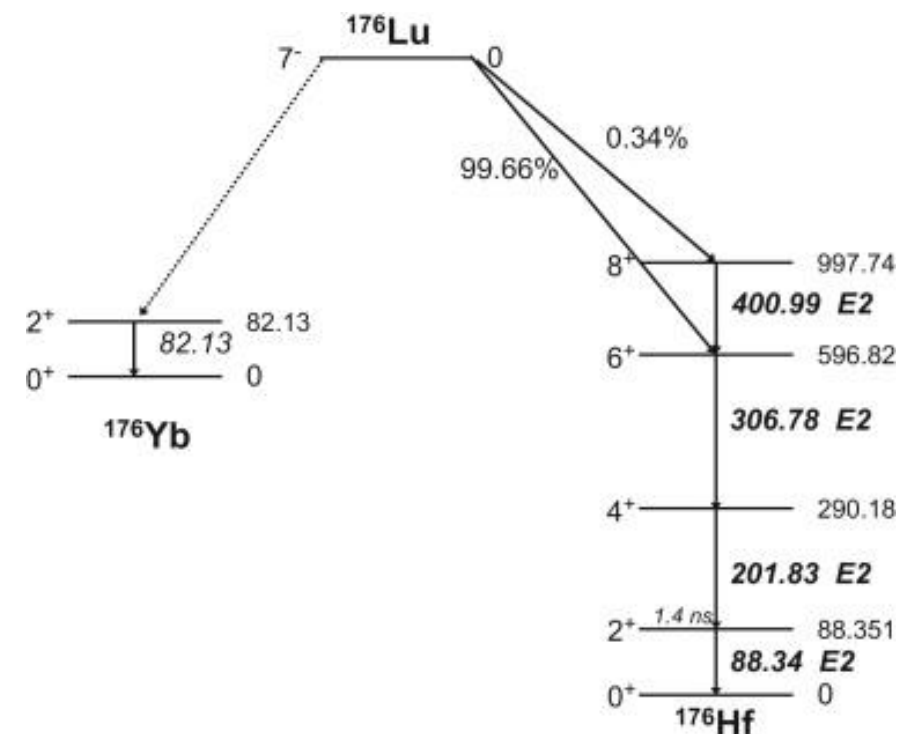


Since the weak force is asymmetric the decay is this “forbidden” as it results in a parity violation and have much longer decay time due to being more rare



Electron Capture Decay

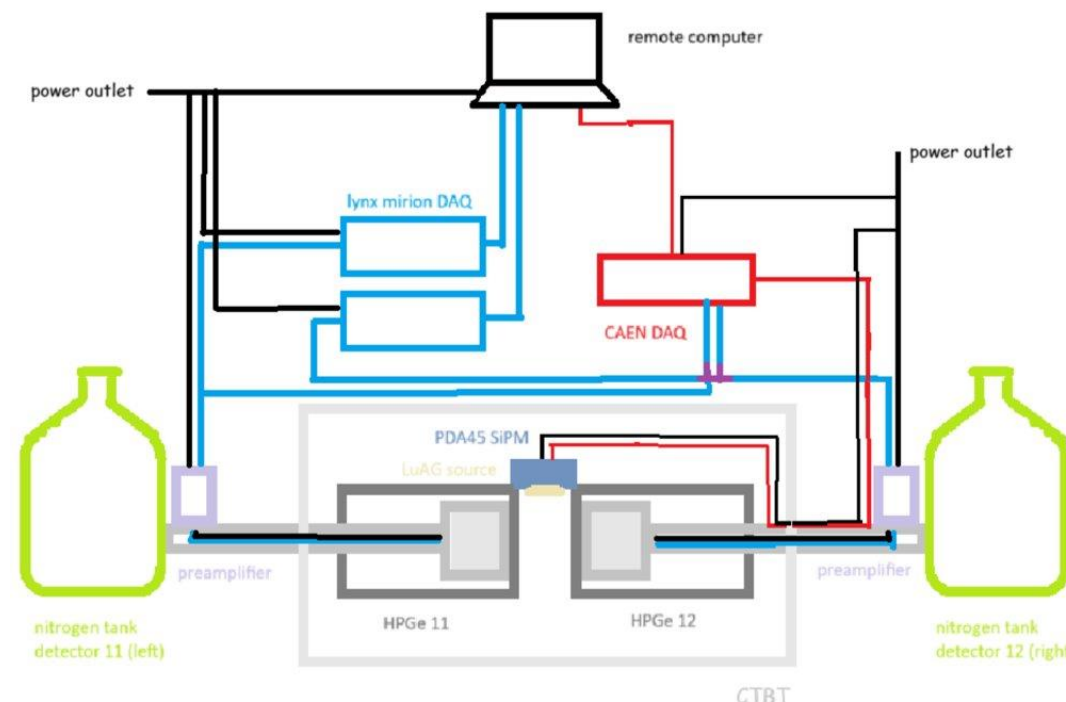
- After the electron capture, the daughter nucleus is left excited, and de-excites via:
 - Atomic de-excitation:
 - Following K-shell capture, characteristic X-rays and/or Auger electrons from Yb (2-7 keV)
 - Nuclear de-excitation
 - Gamma ray @ 82.1 keV from the ^{176}Yb 2^+ state
 82.1 keV ($2^+ \rightarrow 0^+$) primary signature
 Combined with L-shell binding: 92.6 keV peak



Detector Setup

Utilizing a triple coincidence technique

- HPGe Detectors
 - Nuclear de-excitation (gamma rays)
- SiPM
 - Atomic de-excitation causes the LuAG to scintillate light (takes 60-80 ns)

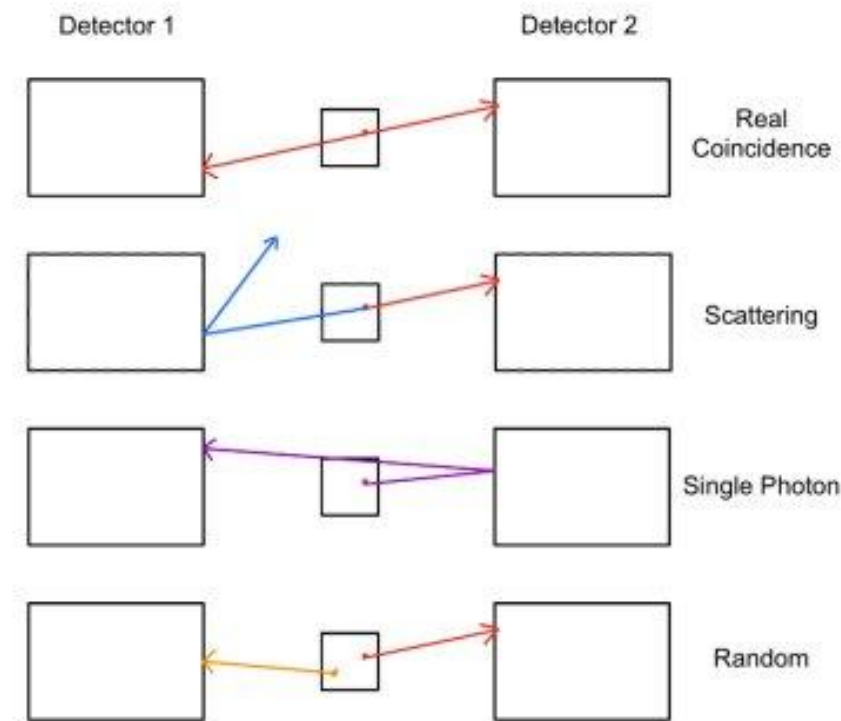


HPGe Detectors

Our HPGe detectors have very high energy range, which helps us measure high energy backgrounds

Measures the gammas emitted by the de-excitation of the Yb-176 after electron capture

Our HPGe detectors are used to detect the two gamma photons which emit 180° from each other



SiPM & Event Signatures

Excited State EC produces:

- **Atomic de-excitation:**
characteristic X-ray and/or Auger electrons

Lu K α X-ray = 54.6 keV

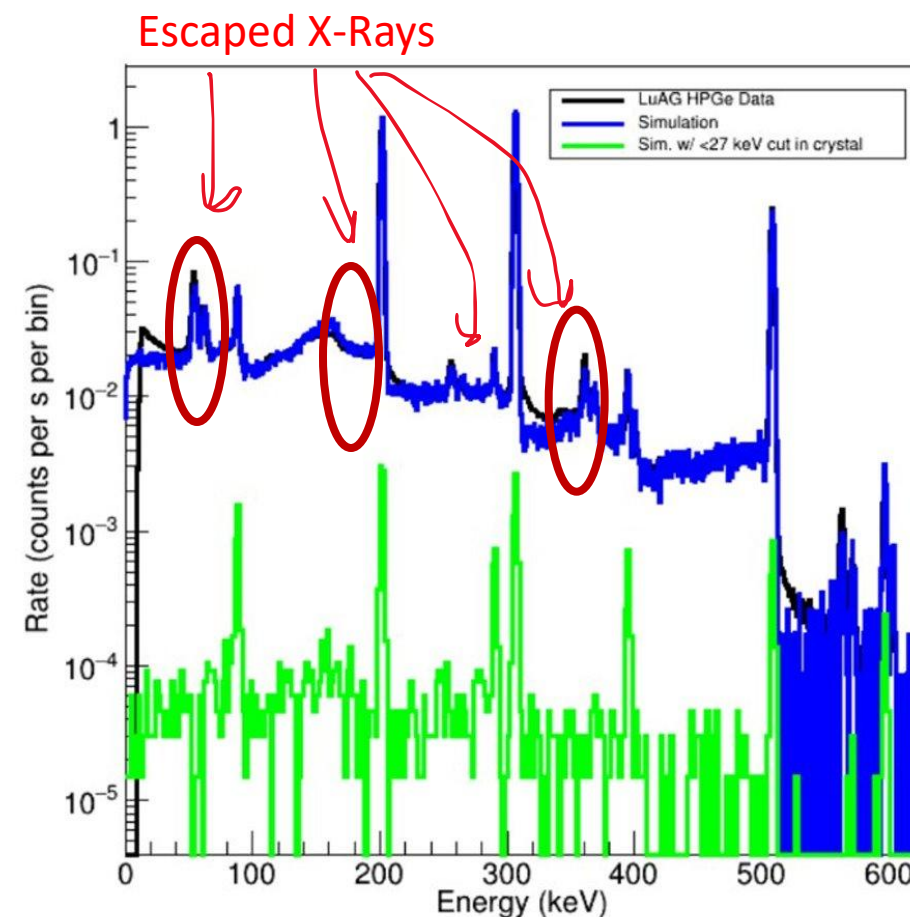
Lu K β X-ray = 61.3 keV

- **Nuclear de-excitation:**

$^{176}\text{Lu} \rightarrow ^{176}\text{Hf} (2^+ \rightarrow 0^+) = 88.3 \text{ keV}$

$^{176}\text{Lu} \rightarrow ^{176}\text{Hf} (4^+ \rightarrow 2^+) = 202 \text{ keV}$

$^{176}\text{Lu} \rightarrow ^{176}\text{Hf} (6^+ \rightarrow 4^+) = 307 \text{ keV}$



Coincidence Technique

RAMPS utilizes the following coincidence technique:

$$\text{EC}^* \text{ event} = [\text{Energy in scintillator}] \otimes [\text{Energy in HPGe}] \otimes [\text{Timing}]$$

Where:

- LuAG crystal scintillator primarily detects the atomic de-excitation which is scintillated, and then detected by the SiPM
- HPGe detects the nuclear de-excitation gamma rays @ high resolution
- Timing coincidence window ensures events are correlated

This ultimately helps us reduce our backgrounds

Data Processing

Primary discrimination method

- By making energy cuts in the scintillator, we can reject β^- events as they deposit significantly more energy into it
- These β^- decay events can be cut to correspond to the 82.1 keV Yb gamma rays, as well as cuts for the atomic x-rays/ auger electrons at 2-7 keV

Moving forward Winter 2026



- Import and integrate (SiPM holder) the new SiPM from Canadian Nuclear Laboratories
- Complete First LuAG Crystal Data Acquisition with the SiPM & HPGe
- Experiment Run for 60 Days?